Continued…
The Threat Environment: Attackers and Their Attacks

Reference:

Corporate & Network Security, Chapter 1

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Agenda

- The threat environment—attackers and their attacks
- Basic security terminology
- Employee and ex-employee threats
- Traditional external attackers
- The criminal era and competitor threats
- Cyberwar and cyberterror

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Rootkits**
    - Take control of the super user account (root, administrator, etc.)
    - Can hide themselves from file system detection
    - Can hide malware from detection
    - Extremely difficult to detect (ordinary antivirus programs find few rootkits)

Rootkit detection programs often are specific to particular rootkits

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Rootkits**

In 2005 Sony BMG downloaded a rootkit onto the PCs of people playing Sony BMG media disks. The discovery of this digital rights management (DRM) rootkit generated extreme negative publicity. The negative publicity increased when it was discovered that the rootkit left the PC open to attack by anyone.

Lemos, http://www.securityfocus.com

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2.3.3 Other Malware Attacks

- **Mobile Code**

When you download a webpage it may contain executable code as well as text, images, sounds, and video. This is called *mobile code* because it executes on whatever machine downloads the webpage. In most cases mobile code is innocent and often is necessary if a user wishes to use a website's functionality.

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Other Malware Attacks

- **Mobile Code**
  - Executable code on a webpage
  - Code is executed automatically when the webpage is downloaded
  - Javascript, Microsoft Active-X controls, etc.
  - Hostile code can do damage if computer has vulnerability

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Other Malware Attacks

- **Social Engineering in Malware**

Social engineering attacks take advantage of flawed human judgment by convincing the victim to take actions that are counter to security policies

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Other Malware Attacks

- **Social Engineering in Malware**
  - Social engineering is attempting to trick users into doing something that goes against security policies

For example if an employee receives an email message warning about a mass layoff being imminent, he or she may open an attachment and therefore download a virus, worm, or trojan horse.

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Other Malware Attacks

- **Social Engineering in Malware**
  - Several types of malware use social engineering
    - Spam
    - Phishing
    - Spear phishing (aimed at individuals or specific groups)
    - Hoaxes

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Other Malware Attacks

- **Social Engineering in Malware**
  - Spam

The bane of all email users is **spam** which is defined as unsolicited commercial e-mail.

In addition to being annoying, spam messages are often fraudulent or advertize dangerous products

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Other Malware Attacks

- **Social Engineering in Malware**
  - Spam

Spam has become a common vehicle for distributing viruses, worms, trojan horses, and many other types of malware

According to MessageLabs, 73% of all e-mail messages were spam in March 2009.
http://www.messagelabs.com

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Other Malware Attacks

- **Social Engineering in Malware**
  - Spam

Even the load on networks caused by simply transmitting and storing spam can be significant. New forms of spam consist of image bodies in stead of text bodies to avoid detection from scanning programs.

Image spam messages are much larger than traditional text spam messages.

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Other Malware Attacks

- **Social Engineering in Malware**
  - Phishing

In phishing attacks victims receive email messages that appear to come from a bank or another firm with which the victim does business. The message may even direct the victim to an authentic-looking website. The official appearance of the message and website often fool the victim into giving out sensitive information.

A Gartner survey in 2007 revealed that US consumers were scammed out of USD 3.2 billion that year.

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Other Malware Attacks

- **Social Engineering in Malware**
  - Phishing

In 2004 when phishing was fairly new but already well known to consumers, a study showed consumers a group of email messages and asked whether each email was a phishing attack or not.

The consumers judged 28% of the phishing messages to be legitimate messages. They also believed a fair number of legitimate messages were phishing messages.

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Other Malware Attacks

- **Social Engineering in Malware**
  - Spear phishing (aimed at individuals or specific groups)

Normally phishing attacks tend to appeal broadly to many people so they can dupe as many people as possible.

In one case a number of CEOs received a message disguising itself as a court order. The message directed the CEOs to a website uscourts.com. There the CEOs could find court documents and a plug-in to view the documents. The plug-in was spyware ! (http://www.pcworld.com) ; 2008

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Other Malware Attacks

- **Social Engineering in Malware**
  - Hoaxes

The sulfnbk.exe hoax told computers that a virus called AOL.exe was travelling around the Internet. The hoax said that they should delete the file sulfnbk.exe. Victims who did so were really deleting their AOL access

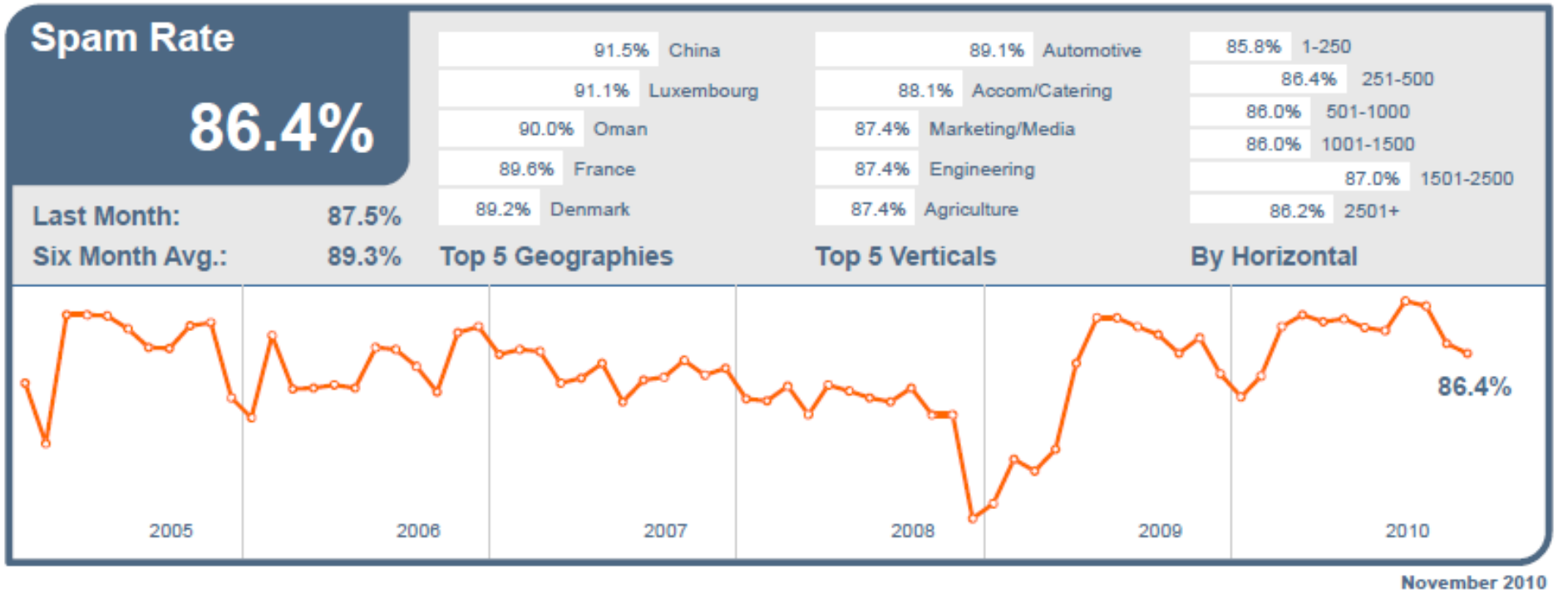
Other hoaxes have tried to persuade victims to delete their antivirus protection and even critical operating files needed for their computers operation.

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MessageLabs Intelligence
http://www.messagelabs.com

by Symantec™

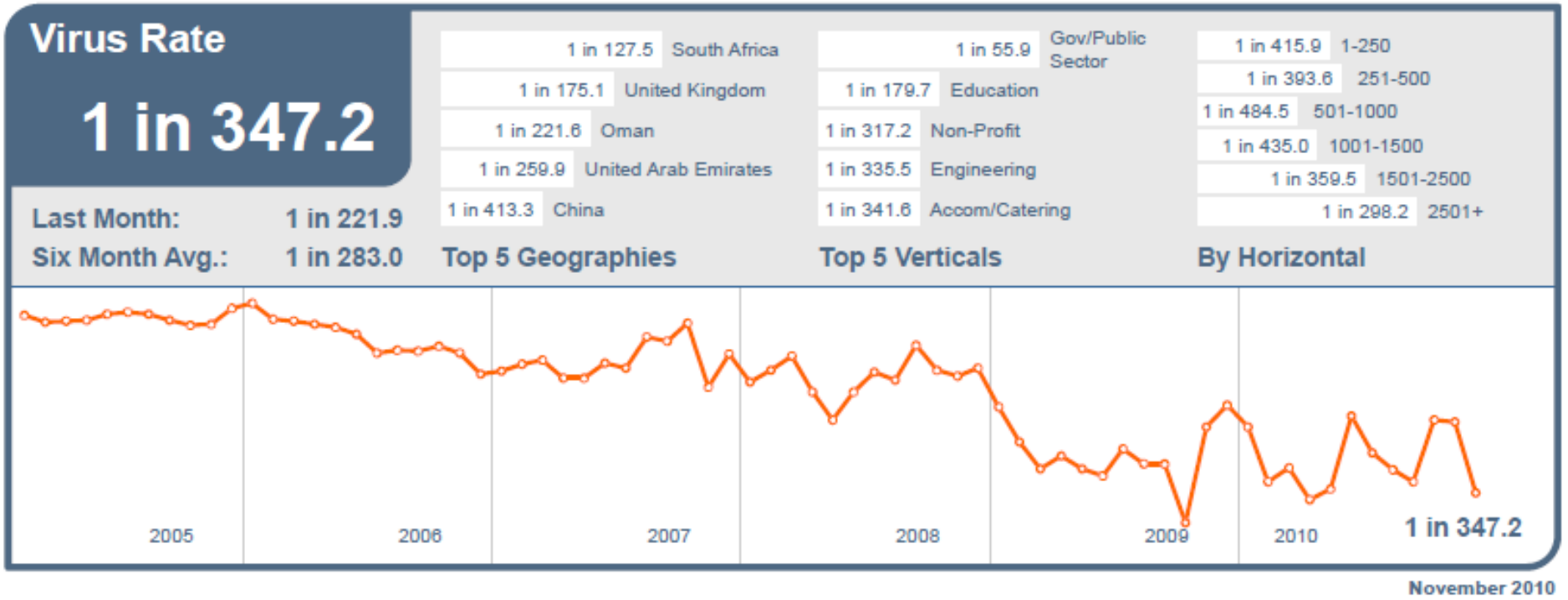
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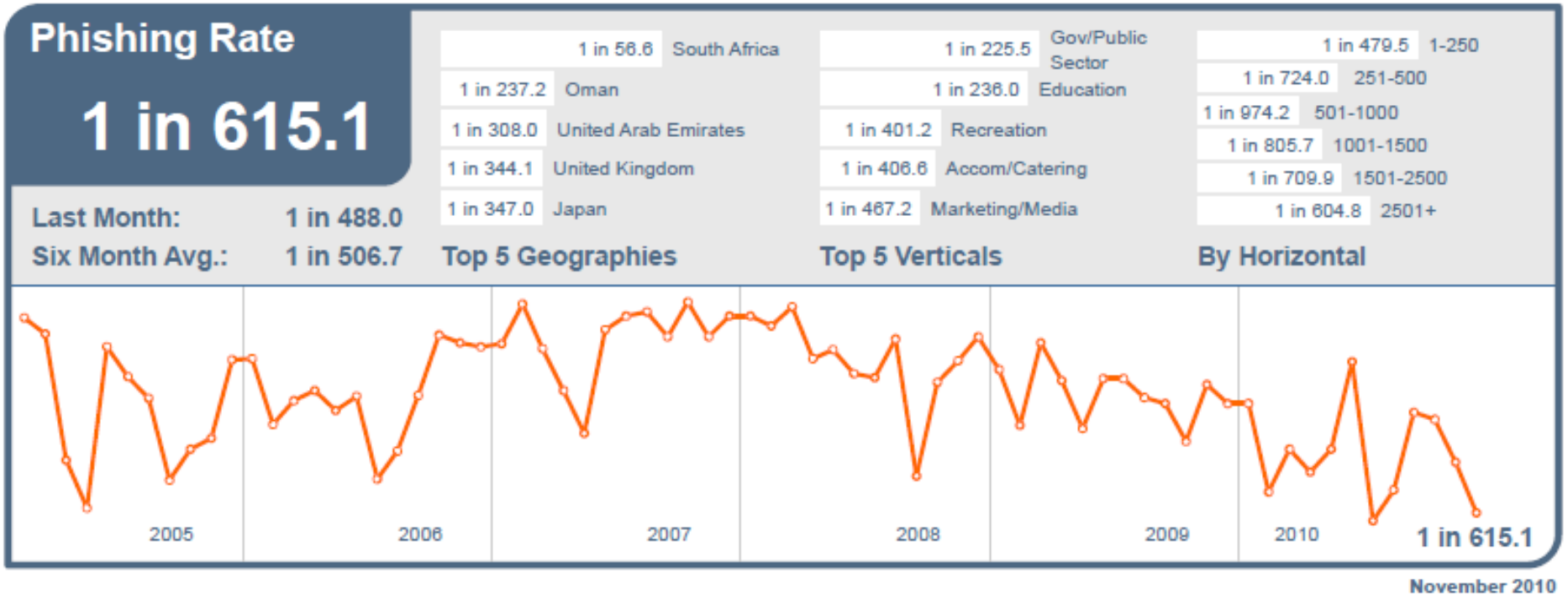
MessageLabs Intelligence
http://www.messagelabs.com



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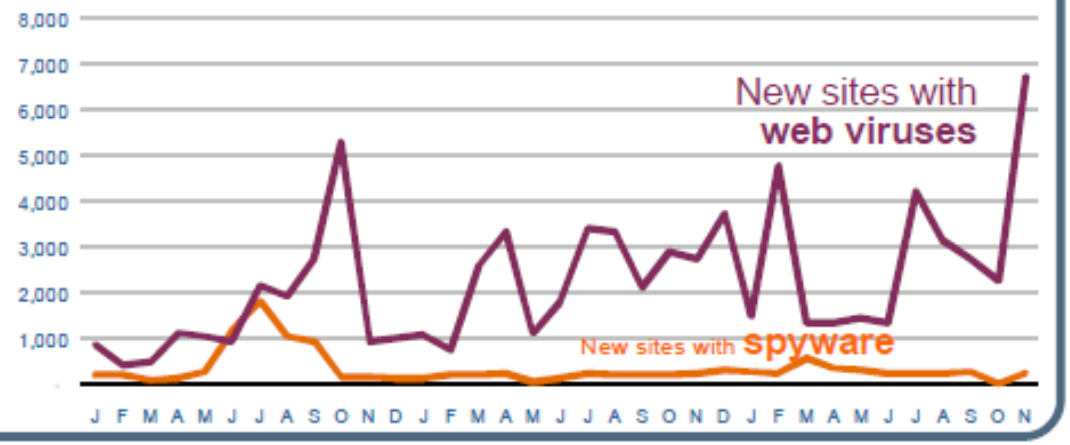
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MessageLabs Intelligence
http://www.messagelabs.com

Web Security Services (Version 2.0) Activity:

New Malware Sites per Day

New sites with **spyware**	235/day
New sites with **web viruses**	6,720/day
Total	**6,955/day**



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McAfee Spam Checklist (Do…)

- Unsubscribe from legitimate mailings that you no longer want to receive.
- Be selective about the Web sites where you register your email address.
- Avoid publishing your email address on the Internet.
- Delete all spam.
- Avoid clicking on suspicious links in email or IM messages as these may be links to spoofed websites.

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McAfee Spam Checklist (Do not…)

- Open unknown email attachments. These attachments could infect your computer.
- Reply to spam. Typically the sender's email address is forged, and replying may only result in more spam.
- Fill out forms in messages that ask for personal or financial information or passwords
- Buy products or services from spam messages.
- Open spam messages.

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